Theoretical and Applied Fracture Mechanics 117 (2022) 103203

Contents lists available at ScienceDirect

Theoretical and Applied Fracture Mechanics

journal homepage: www.elsevier.com/locate/tafmec

Modeling crack propagation in heterogeneous granite using grain-based phase field method

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ARTICLE INFO

Keywords:
Grain-based phase field method (GB-PFM)
Granite
Rock heterogeneity
Cracking behavior
Intragrain cracks

ABSTRACT

The cracking behavior and mechanical properties of brittle rocks is controlled by its microstructure. In this work, an extended grain-based phase field method (GB-PFM) based on Voronoi tessellation technology is proposed to investigate the mechanical properties and crack propagation in heterogeneous brittle rocks by considering material heterogeneities and microstructure heterogeneities at the grain scale. Then, a trial and error calibration procedure is developed after comprehensive study of uncertain parameters such as grain boundaries thickness and critical energy release rate. It is shown that greater grain boundaries thickness results in a lower tensile strength and lower Young's modulus, and greater number of intergrain cracks. Then, a group of models with single pre-existing flaw (flaw inclination angles: 0°, 30°, 45°, 60°, 90°) are built according to calibration parameters to study the mechanical properties and cracking behavior of heterogeneous granite under direct tension. The path of the macrocracks obtained from GB-PFM shows a more noticeable tortuous nature as compared with that results by others phase field methods. The variation of flaw inclination has an obviously influence on crack propagation. Failure patterns obtained by GB-PFM revealing the macrocracks are composed of intragrain and intergrain cracks at the microscopic view. Moreover, the stress field distribution obtained by GB-PFM is agree well with the theoretical analyses and other numerical simulations. The above simulated results prove that the GB-PFM provides an applicable numerical method for efficiently reproducing the microstructure and the related crack propagation of brittle crystalline rocks, which greatly expand the application of the phase field methods.

1. Introduction

Granite is a typical brittle rock material with damage evolution and crack propagation controlled mainly by both the microstructure (including mineral composition and content, mineral grains size and shape, and grain boundaries networks) and mechanical properties (including grain-interior properties and grain boundaries properties) [4,77,85,54,60]. A lot of structures, like dams, bridges, tunnels, have been constructed with granite [7]. Therefore, exploring the damage evolution and crack propagation of granite has great theoretical and practical significance.

To date, many researchers have studied the crack initiation, propagation and coalescence mechanisms of rocks or rock-like materials through laboratory tests [32,46,53,83]. The results of these laboratory tests are of great help in understanding the crack propagation mechanism and damage evolution in brittle rocks. Although different laboratory techniques (e.g., acoustic emission [93], scanning electron microscopy [42]) have been used to study the cracking behavior of brittle rocks, but the damage evolution and crack propagation is still far from fully understood, especially, when take account of the microstructure of brittle rocks [55]. In addition, there are several other shortcomings for laboratory tests: (a) it is limited in quantitative description of crack propagation progress inside of the mineral grains; (b) the laboratory tests are generally expensive, time consuming and poor repeatability.

Alternatively, numerical simulation have been widely used for mechanical investigation of rocks or rock-like materials in recent years. Many traditional numerical methods such as finite element method [70,25], finite difference method [10,62], and discrete element method [12,6] have been used. Recently, many new numerical methods such as

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https://doi.org/10.1016/j.tafmec.2021.103203
Received 7 October 2021; Received in revised form 29 November 2021; Accepted 29 November 2021
Available online 8 December 2021
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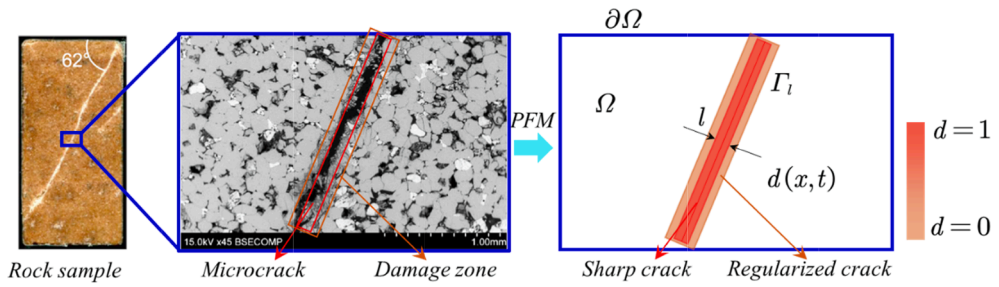


Fig. 1. Laboratory observation of the damage zone in brittle rocks (modified from [43,87] and the regularization of sharp crack topology.

smoothed particle hydrodynamics [88], numerical manifold method [81] and peridynamics [76] have shown many advantages of modeling cracking and failure in brittle rocks. For example, peridynamics can remove the crack-tip stress singularity [5].

In recent years, some researches have proposed a new numerical method, the phase field method (PFM), which is based on classical Griffith theory [21]. All the contributions in PFMs [37,91,51,58,30,87] have shown the potential of a PFM for simulating initiation and propagation of microcracks. As one of the advantages, the PFM does not need to set the shape and size of the propagating crack in advance, and can better describe the processes of crack initiation, propagation and coalescence [95]. This provides a rigorous and accurate theoretical framework and numerical simulation method for the simulation of brittle fracture. At present, many researchers have used the PFM to simulate the crack propagation and failure processes of various types of rocks or rock-like materials under different loading conditions, such as gypsum [91], concrete [14,84,89,73,27], and rocks [67,58,74].

However, brittle rocks contain different types of heterogeneities. As an example, at microscale, granite is heterogeneous multiphasic materials, which primarily consists of K-feldspar, plagioclase, quartz, and biotite. The microcracks initiation, propagation and coalescence in granite are inherently influenced by the presence of such heterogeneities. Rock heterogeneities is induced by the distribution of mineral grains and their mechanical properties at the grain scale, e.g., Poisson's ratio and critical energy release rate, etc. [45,24]. And rock heterogeneities also exists within single mineral grain [15,59]. For example, for quartz grain, the critical energy release rate varies along different crystallographic directions [59]. In such heterogeneous multigrain systems, the minerals grains may exhibit preferential cleavage planes arising due to crystallographic direction-dependent material properties. Such as, plagioclase shows cleavage planes in both [0 0 1] and [0 1 0] and microcracks usually followed the cleavage planes [31]. In addition, the grain boundaries in brittle rocks exists as a weak plane of micromechanical properties.

To better simulate the microstructure of crystalline rocks, Potyondy [57] proposed a grain-based method (GBM), which is based on Particle Flow Code [28]. The GBM realized by numerical simulation has become an extremely important method for studying brittle rock microstructure. The GBM concept is gradually extended to other numerical simulation methods, and the grain boundaries is no longer confined to the conventional Voronoi polygon, and gradually forms a transition from regular boundary to irregular boundary, thus forms a variety of numerical simulation methods, such as GBM-UDEC [33], GB-FDEM [96,8], GBM-PFC [92,24], *etc.*

In the mentioned works based on PFMs, many researches only deterministic geometries and crack paths were considered [94,87,58,18]. Since microstructure randomness strongly impacts the macroscopic response of (quasi-) brittle rocks in various ways, ranging from size effects [71,72] to failure patterns and ultimate mechanical properties [33,54,92,35,24], the development of approaches incorporating Voronoi tessellation and PFM together is a natural path to extend the predictive capabilities in brittle fracture simulations. The main objective of this work is to propose an extended grain-based phase field method (GB-PFM) to model crack propagation in heterogeneous brittle rocks, like granite. In the proposed framework, the microstructure of brittle rocks is specifically defined through Voronoi tessellation. This particular setting ensures that material heterogeneities and microstructure heterogeneities at the grain scale and allows for the crack propagation at the microscopic level.

In this paper, an extend GB-PFM is proposed to investigate the mechanical properties and cracking behavior of Lac du Bonnet (LdB) granite [46], a typical brittle crystalline rock, under direct tension. The main structure of the paper is as follows. In Section 2, the PFM is introduced. In Section 3, the procedure to construct a GB-PFM model based on Voronoi tessellation is introduced. In Section 4, sensitivity analysis of preferred cleavage orientation, the grain boundary thickness and critical energy release rate, and their effects on mechanical properties and cracking behavior are investigated and discussed. In Section 5, cracking behavior of LdB granite with single pre-existing flaw under direct tension test is studied and compared with laboratory results. In Section 6, several limitations of GB-PFM and research trends in the future are discussed. And conclusions are given in Section 7.

2. Phase field modeling of brittle fracture

2.1. Phase field approximation of the fracture energy

Let $\Omega\subset\mathbb{R}$ be a limited domain with an external boundary $\partial\Omega$, which contains several cracks Γ, see Fig. 1.

It is proposed to approximate the total area of cracks A_Γ by a crack surface density function Γ_d defined as follows:

Consider a cracked solid with the strain energy density $\psi(\boldsymbol{\varepsilon})$. In the absence of the external forces, the variation of the total free energy Π due to an incremental increase in the crack area dA is given by

$$\frac{\mathrm{d}\Pi}{\mathrm{dA}}=\frac{\mathrm{d}\psi(\boldsymbol{\varepsilon})}{\mathrm{dA}}+\frac{\mathrm{d}W_\mathrm{c}}{\mathrm{dA}}=\frac{\mathrm{d}\psi(\boldsymbol{\varepsilon})}{\mathrm{dA}}+G_\mathrm{c}=0 \tag{1}$$

where $\boldsymbol{\varepsilon}$ is the strain tensor and W_c is the work required to create new cracks in domain Ω. And G_c is the critical energy release rate, which is a material property that characterizes the crack resistance. Under the plane strain assumption, the theoretic value of G_c^0 is computed as $G_\mathrm{c}^0=K_{Ic}^2(1-\nu^2)/E$, where K_{Ic}^2, ν and E are the fracture toughness, Poisson's ratio and Young's modulus, respectively. Griffith's energy balance can be formulated in an integral form of *Eq.* (1) as follows

$$\Pi(\boldsymbol{u},\Gamma)=\Pi_b(\boldsymbol{u},\Gamma)+\Pi_s(\Gamma)=\int_\Omega\psi(\boldsymbol{\varepsilon})\mathrm{d}V+\int_\Gamma G_c\mathrm{d}\Gamma \tag{2}$$

where Π_b is the bulk energy, Π_s is the surface energy,$\boldsymbol{u}$ is the displacement field and V is the volume of the solid. The strain tensor $\boldsymbol{\varepsilon}$ can be formulated as follows

$$\varepsilon_{ij}=\frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right)\qquad(i,j=1,2,3) \tag{3}$$

An auxiliary scalar $d\in[0,1]$ is the phase field variable (0: undamaged state; 1: fully damaged) can be used to track the crack surface (as

shown in Fig. 1). The crack surface density function per unit volume is introduced by Ambrosio and Tortorelli [2] and is used here

$$\Gamma_d(d,\nabla d)=\frac{d^2}{2l}+\frac{l}{2}|\nabla d|^2 \tag{4}$$

where l is a length scale parameter, which characterizes the width of the smeared crack, as shown in Fig. 1.

In the framework of continuous damage mechanics, a degradation of materials by microcracks is characterized by the reduction of elastic strain energy stored in solids. The most widely used form for the degradation function $g(d)$ is $g(d)=(1-d)^2+k$[49], where k is a parameter chosen to be as small as possible, and is set to 10^{-9}. The total free energy functional Eq. (2) can be rewritten as

$$\Pi=\int_\Omega\left\{\left[(1-d)^2+k\right]\psi_0(\boldsymbol{\varepsilon})+G_c\left(\frac{d^2}{2l}+\frac{l}{2}|\nabla d|^2\right)\right\}\mathrm{d}V \tag{5}$$

where ψ_0 denotes the strain energy density of the undamaged solid.

In order to better mimic the preferential cleavage planes in different minerals, a symmetric second-order tensor $\boldsymbol{\omega}$ is implemented in *Eq.* (4) as follows

$$\Gamma_d(d,\nabla d,\boldsymbol{\omega})=\frac{d^2}{2l}+\frac{l}{2}(\nabla d\cdot\boldsymbol{\omega}\cdot\nabla d) \tag{6}$$

the $\boldsymbol{\omega}$ describing the preferred cleavage plane in minerals is given by [11]

$$\boldsymbol{\omega}=\mathbf{1}+\beta(\mathbf{1}-\boldsymbol{M}\otimes\boldsymbol{M}) \tag{7}$$

where $\mathbf{1}$ is the identity tensor, β is a penalty parameter, $\mathbf{M}$ is a unit vector normal to the preferred cleavage plane, and $\otimes$ denotes the tensor product of two vectors. Setting $\beta\gg1$ greatly increases surface energy of fracture on planes not normal to $\mathbf{M}$; setting $\beta=0$ results in isotropic damage (see Nguyen et al. [51] for more details on this aspect). Here, we consider $\beta=10$ in this work [52]. Therefore, *Eq.* (7) can be expressed as

$$\boldsymbol{\omega}=\begin{bmatrix}1+\beta sin^2(\theta) & -\beta cos(\theta)sin(\theta)\\ -\beta cos(\theta)sin(\theta) & 1+\beta cos^2(\theta)\end{bmatrix};\qquad \boldsymbol{M}=\begin{bmatrix}cos(\theta)\\ sin(\theta)\end{bmatrix} \tag{8}$$

where θ is the preferred cleavage orientation, and $\theta\in[-90°,90°]$. It is should be mentioned that within the grain boundaries, the existence of the preferred cleavage plane is not considered in this work. Furthermore, the preferred cleavage orientation θ in different minerals are randomly generated.

Therefore, *Eq.* (5) is rewritten as

$$\Pi=\int_\Omega\left\{\left[(1-d)^2+k\right]\psi_0(\boldsymbol{\varepsilon})+G_c\left[\frac{d^2}{2l}+\frac{l}{2}(\nabla d\cdot\boldsymbol{\omega}\cdot\nabla d)\right]\right\}\mathrm{d}V \tag{9}$$

To avoid unphysical compression fracture in anisotropic solids, the strain energy density can be decomposed as

$$\psi(\boldsymbol{\varepsilon},d)=\left[(1-d)^2+k\right]\psi_0^+(\boldsymbol{\varepsilon})+\psi_0^-(\boldsymbol{\varepsilon}) \tag{10}$$

where

$$\psi_0^\pm(\boldsymbol{\varepsilon})=\frac{\lambda}{2}\langle tr(\boldsymbol{\varepsilon})\rangle_\pm^2+\mu tr\left[(\boldsymbol{\varepsilon}_\pm)^2\right] \tag{11}$$

where $tr(\cdot)$ denotes the trace of the tensor, $\langle x\rangle_\pm=\frac{1}{2}(x\pm|x|)$.$\lambda$ and μ are the Lamé constants defined in 2D as

$$\lambda=\frac{\nu E}{(1+\nu)(1-2\nu)},\mu=\frac{E}{2(1+\nu)} \tag{12}$$

wherein only the positive part $\psi_0^+(\boldsymbol{\varepsilon})$ due to tension contributes to the phase field evolution.

The strain tensor $\boldsymbol{\varepsilon}$ is given in the form of spectral decomposition as

$$\boldsymbol{\varepsilon}_\pm=\sum_{I=1}^{3}\langle\varepsilon_I\rangle_\pm\boldsymbol{n}_I\otimes\boldsymbol{n}_I\quad\text{and}\quad I=1,2,3 \tag{13}$$

where ε_I,$\boldsymbol{n}_I$ are the principal strains, principal strain directions, respectively.

The stress tensor can be computed by

$$\boldsymbol{\sigma}=\frac{\partial\psi(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}}=\left[(1-d)^2+k\right]\frac{\partial\psi_0^+(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}}+\frac{\partial\psi_0^-(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}} \tag{14}$$

which is used to derive the elastic fourth-order tensor $\mathbb{C}$ for anisotropic solids

$$\begin{aligned}\mathbb{C}=\frac{\partial\boldsymbol{\sigma}}{\partial\boldsymbol{\varepsilon}}&=\frac{\partial}{\partial\boldsymbol{\varepsilon}}\left\{\left[(1-d)^2+k\right]\frac{\partial\psi_0^+(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}}+\frac{\partial\psi_0^-(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}}\right\}\\&=\left[(1-d)^2+k\right]\{\lambda H[tr(\boldsymbol{\varepsilon})\mathbb{J}+2\mu]\mathbb{P}^+\}\\&\quad+\{\lambda H[-tr(\boldsymbol{\varepsilon})\mathbb{J}]+2\mu\mathbb{P}^-\}\end{aligned} \tag{15}$$

where $H(x)$ is the Heaviside function. $\mathbb{J}$ is a fourth-order tensor such that if $i=k$ and $j=l$, the result is unity, and if one of the equalities is violated, then the result is zero. Projection tensors $\mathbb{P}^\pm$ can be expressed based on

$$\mathbb{P}^\pm=\sum_{\mathrm{a}}^{\mathrm{b}}\left(\mathrm{H}(\pm\varepsilon_{\mathrm{a}})\mathbb{Q}\ominus_{\mathrm{a}}+\sum_{\mathrm{b},\mathrm{b}\neq\mathrm{a}}^{\mathrm{d}}\frac{\langle\varepsilon_{\mathrm{a}}\rangle_\pm}{2(\varepsilon_{\mathrm{a}}-\varepsilon_{\mathrm{b}})}(\mathbb{G}_{\mathrm{ab}}+\mathbb{G}_{\mathrm{ba}}\ominus)\right) \tag{16}$$

with

$$\begin{aligned}\boldsymbol{M}_a&=\boldsymbol{n}_a\otimes\boldsymbol{n}_a\\(\mathbb{Q}_{\mathrm{a}})_{ijkl}&=(\boldsymbol{M}_a)_{ij}(\boldsymbol{M}_a)_{kl}\\(\mathbb{Q}_{\mathrm{ab}})_{ijkl}&=(\boldsymbol{M}_a)_{ik}(\boldsymbol{M}_b)_{jl}+(\boldsymbol{M}_a)_{il}(\boldsymbol{M}_b)_{jk}\end{aligned} \tag{17}$$

Their detailed expressions can be referred to Liu et al. [37] and Miehe and Lambrecht [48].

Therefore, total free energy is expressed as

$$\Pi=\int_\Omega\left\{\left[(1-d)^2+k\right]\psi_0^+(\boldsymbol{\varepsilon})+\psi_0^-(\boldsymbol{\varepsilon})+G_c\left[\frac{d^2}{2l}+\frac{l}{2}(\nabla d\cdot\boldsymbol{\omega}\cdot\nabla d)\right]\right\}\mathrm{d}V \tag{18}$$

2.2. Derivation of the governing equations

The total potential energy functional is then given by

$$\Psi(\boldsymbol{u},d)=W_{\mathrm{int}}(\boldsymbol{u},d)+W_{\mathrm{s}}(d)-W_{\mathrm{ext}}(\boldsymbol{u}) \tag{19}$$

where W_{int}, W_{ext} are the internal and external energy, respectively. The governing balance equations are obtained from the first derivation of *Eq.* (19) as follows

$$\delta\Psi=\delta W_{int}+\delta W_s-\delta W_{ext}=0 \tag{20}$$

The variation of the crack surface can be computed form Eq. (9) as

$$\delta W_s=\int_\Omega G_c\left(\frac{1}{l}d\delta d+l\nabla d\cdot\omega\cdot\nabla\delta d\right)d\Omega \tag{21}$$

The variation of W_{ext} can be written as follows

$$\delta W_{ext}=\int_\Omega b\cdot\delta u d\Omega+\int_{\partial\Omega}h\cdot\delta u\delta\Omega \tag{22}$$

where $\mathbf{b}$ is the body force while $\mathbf{h}$ is the boundary traction field on an external boundary $\partial\Omega$.

Furthermore, the first variation of the internal potential energy in Eq. (9) is given by

$$\begin{aligned}\delta W_{int}&=\int_\Omega\left\{\left[(1-d)^2+k\right]\sigma^++\sigma^-\right\}\delta\varepsilon d\Omega-\int_\Omega 2(1-d)\psi^+d\delta d d\Omega\\&=\int_\Omega\sigma\delta\varepsilon d\Omega-\int_\Omega 2(1-d)\psi^+d\delta d d\Omega\end{aligned} \tag{23}$$

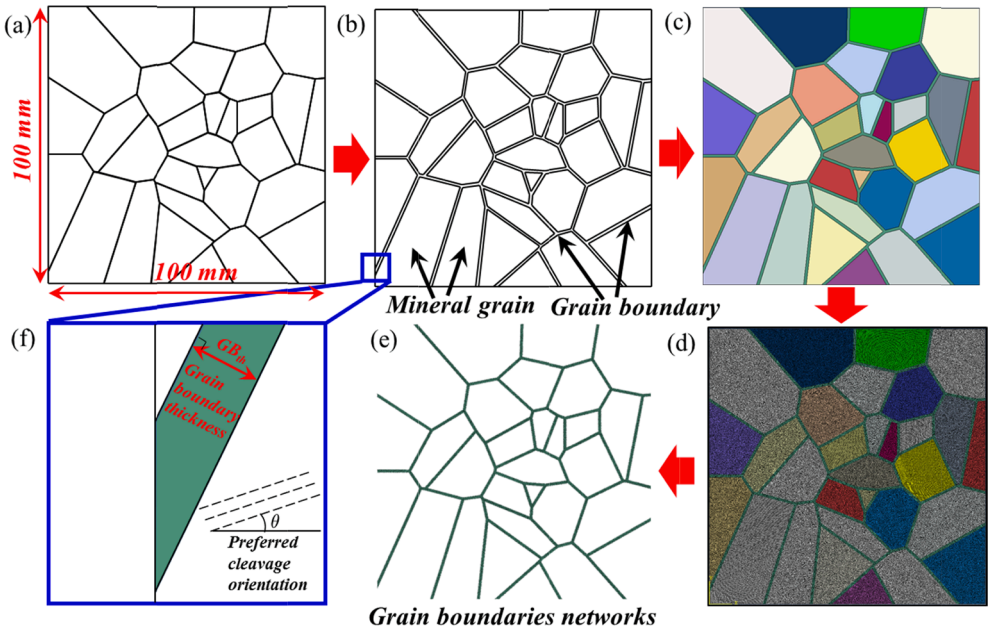


Fig. 2. Generation of the GB-PFM containing 30 grains. **a** Voronoi tessellations generated by Neper [61]. **b** The microstructure containing grain boundaries formed by offsetting each Voronoi unit cell by grain boundary thickness GB_{th}. **c** The microstructure in Abaqus by imported **b**. The colors indicate different mineral grains or grain boundaries (lime green). **d** Finite element meshes of **c**. **e** Grain boundaries networks. And **f** schematic diagram of grain boundary thickness GB_{th} and preferred cleavage orientation θ. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

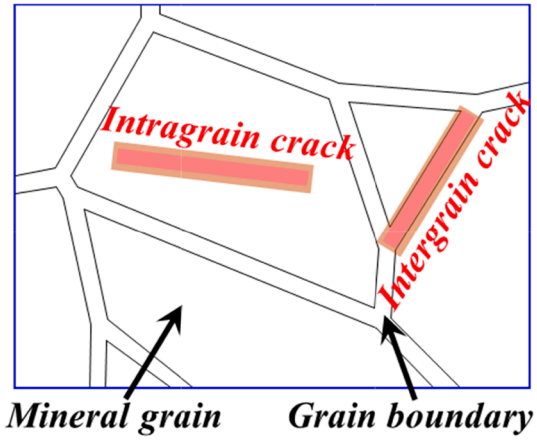


Fig. 3. Schematics illustrating two types of cracking in brittle rocks (intragrain cracks and intergrain cracks).

where the tensile $\boldsymbol{\sigma}^+$ and compressive $\boldsymbol{\sigma}^-$ parts of the Cauchy stresses are given as [50]

$$\boldsymbol{\sigma}^+ = \frac{\partial \psi_0^+(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}}; \quad \boldsymbol{\sigma}^- = \frac{\partial \psi_0^-(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}} \tag{24}$$

To avoid crack healing under loading and unloading histories, we define the crack driving force using the history variable $\mathscr{H}$

$$\mathscr{H} = \max_{\tau \in [0,t]} \left(\psi_0^+(\tau) \right) \tag{25}$$

Table 1
Microproperties of the minerals [29,3,33,44].

Mineral type	E (GPa)	ν	K_{Ic} (MPa$\sqrt{\text{m}}$)	G^0_c (N/mm)	Mean grain size (mm)	Content (%)
K-feldspar	69.8	0.28	4.18	0.231	3.00	40
Plagioclase	88.1	0.26	4.18	0.185	3.50	20
Quartz	94.5	0.08	8.68	0.792	1.50	30
Biotite	33.8	0.36	3.21	0.265	0.75	10

Note: K_{Ic}: Data from Mahabadi [44],G^0_c: $G^0_c = K^2_{Ic}(1-\nu^2)/E$.

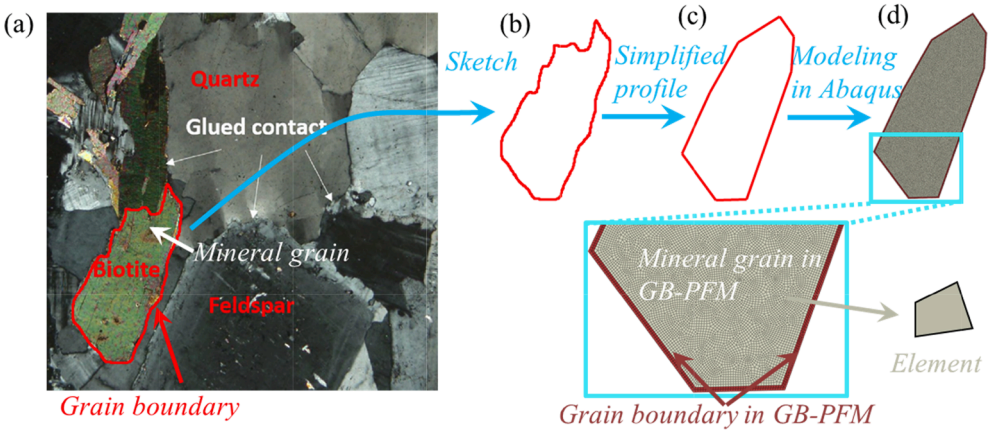


Fig. 4. Generation procedure of GB-PFM from an actual LaB granite. **a** The micrograph of the LdB granite [60], **b** Sketch of the biotite in **a**; **c** Generalized grain boundary profile of **b**; **d** Mineral grain and grain boundary in GB-PFM according to **c**.

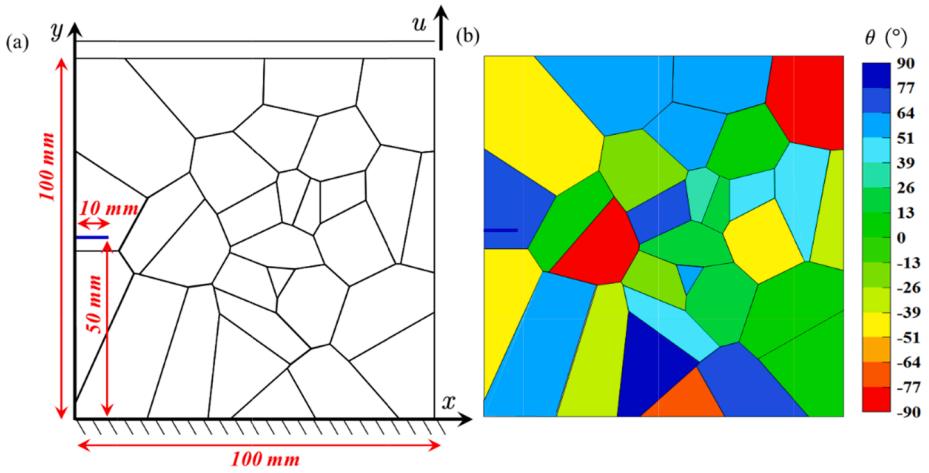


Fig. 5. Model of the GB-PFM containing 30 grains. **a** Boundary conditions; **b** Preferred cleavage orientation. The grain boundarie thickness is not marked.

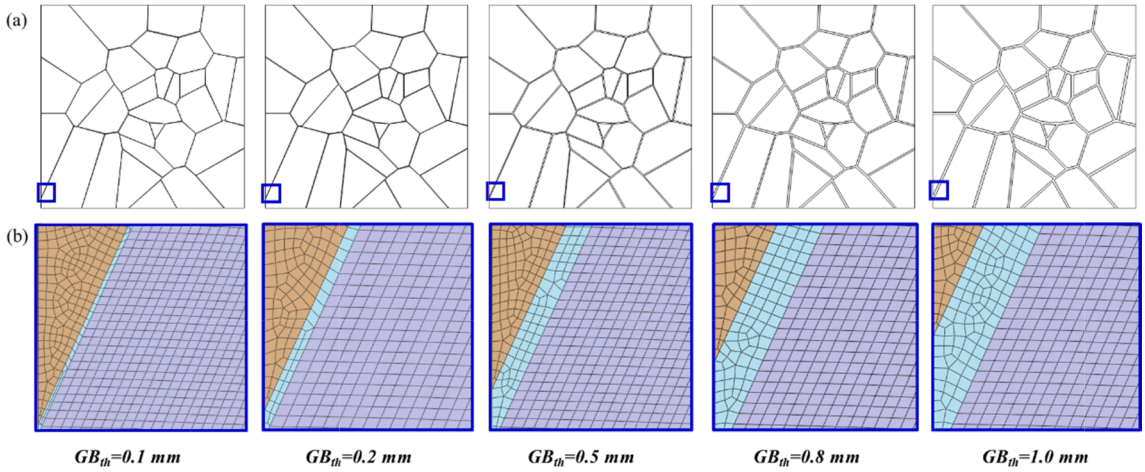


Fig. 6. Numerical specimens with different grain boundaries thickness. **a** Geometry and **b** Zoom-in figures showing grain boundaries (soft blue) and grains (light grayish blue and slightly orange). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where t is the current time. This is the same approach taken in Miehe et al. [49]. For a discussion and a more detailed description of the use of Eq. (25), the reader is invited to refer to the mentioned reference.

The strong form of the governing balance equations of the coupled problem is given as follows

$$\begin{aligned} G_c\left[\frac{1}{l}d - l\Delta d\cdot\boldsymbol{\omega}\right] - 2(1-d)\mathscr{H} = 0 \\ \mathrm{Div}[\boldsymbol{\sigma}] + \boldsymbol{b} = \boldsymbol{0} \end{aligned} \tag{26}$$

Along with the boundary conditions

$$\begin{aligned} \sigma\cdot n = h \quad on \quad \partial\Omega_h \\ \nabla d\cdot n = 0 \quad on \quad \partial\Omega \end{aligned} \tag{27}$$

where **n** denotes the outward unit vector normal to the surface $\partial\Omega$.

2.3. Finite element implementation

In order to obtain numerical solutions of the coupled system of partial differential *Eq.* (26) using finite element method, the weak form of *Eq.* (26) is given as follows

$$\begin{aligned} \int_\Omega (\sigma\delta\varepsilon - b\cdot\delta u)dV - \int_{\partial\Omega_h} h\cdot\delta u dA = 0 \\ \int_\Omega [-2(1-d)\delta d\mathscr{H} + G_c(\frac{1}{l}d\delta d + l\nabla d\cdot\omega\cdot\nabla\delta d)]dV = 0 \end{aligned} \tag{28}$$

The displacement field **u**, the phase field variable d and the corresponding derivatives can be discretized as

$$\begin{aligned} u = \sum_{i=1}^m N_i^u u_i, d = \sum_{i=1}^m N_i d_i and N_i^u = \begin{bmatrix} N_i & 0 \\ 0 & N_i \end{bmatrix} \\ \varepsilon = \sum_{i=1}^m B_i^u u_i and \nabla d = \sum_{i=1}^m B_i^d d_i \end{aligned} \tag{29}$$

where $\boldsymbol{N_i^u}$ is the shape function matrix, N_i denotes the shape function associated with node i, and m is the total number of nodes per element. d_i and $\boldsymbol{u}_i$ are the phase field values and the displacement at node i. The strain matrices are expressed as follows

$$\boldsymbol{B_i^u} = \begin{bmatrix} N_{i,x} & 0 \\ 0 & N_{i,y} \\ N_{i,y} & N_{i,x} \end{bmatrix} \quad \text{and} \quad \boldsymbol{B_i^d} = \begin{bmatrix} N_{i,x} \\ N_{i,y} \end{bmatrix} \tag{30}$$

where $N_{i,x}$ and $N_{i,y}$ are the derivatives of the corresponding shape function with respect to x and y, respectively.

According to Eqs. (28)–(30), the residual with respect to the

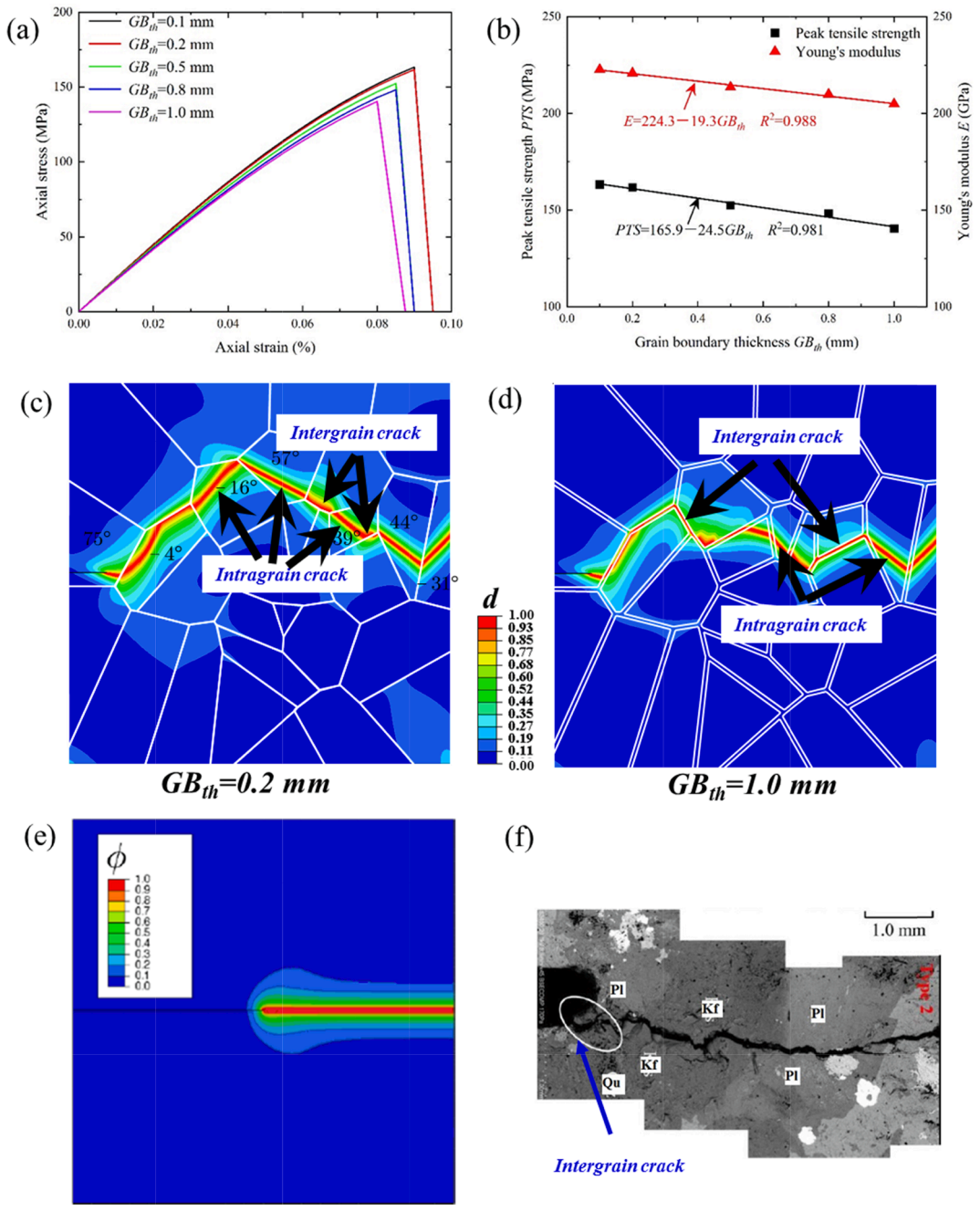


Fig. 7. Macromechanical and macroscopic tensile crack responses of specimens with different grain boundaries thickness and crack patterns observed in laboratory tests. **a** Stress–strain curves of five models; **b** Peak tensile strength and Young's modulus of five models; **c** Crack patterns of specimen GB_{th} = 0.2 mm, and **d** Crack patterns of specimen GB_{th} = 1.0 mm. Note: 75° represents the preferred cleavage plane orientation of the grain; **e** Crack patterns for a homogeneous numerical specimen [30], **f** Crack patterns of Kowloon granite with a pre-existing notch observed in laboratory tests [92].

evolution of the displacement field and phase field can be expressed as

$$\begin{cases} R_i^{\boldsymbol{u}} = \int_{\Omega} \left(\boldsymbol{B}_i^{\boldsymbol{u}}\right)^{\mathrm{T}} \boldsymbol{\sigma} \mathrm{d}V - \int_{\Omega} \left(\boldsymbol{N}_i^{\boldsymbol{u}}\right)^{\mathrm{T}} \boldsymbol{b} \mathrm{d}V - \int_{\partial\Omega_h} \left(\boldsymbol{N}_i^{\boldsymbol{u}}\right)^{\mathrm{T}} \boldsymbol{h} \mathrm{d}A \\ R_i^{d} = \int_{\Omega} \left\{ -2(1-d)N_i \mathscr{H} + \mathrm{G_c} \left[\frac{1}{l} \mathrm{N_i d} + l \left(\mathbf{B}_{\mathrm{i}}^{\mathrm{d}}\right)^{\mathrm{T}} \cdot \boldsymbol{\omega} \cdot \nabla \mathrm{d} \right] \right\} \mathrm{d}V \end{cases} \tag{31}$$

In order to obtain the solutions for which $\boldsymbol{R}^{\boldsymbol{u}} = \mathbf{0}$ and $\boldsymbol{R}^{d} = \mathbf{0}$, a quasi-Newton solution method is used, which is a built-in feature in Abaqus and the stiffness matrix is updated in each iteration following the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm [30]. The initial guess for the system takes the following form

$$\begin{Bmatrix} u \\ d \end{Bmatrix}_{t+\Delta t} = \begin{Bmatrix} u \\ d \end{Bmatrix}_{t} - \begin{bmatrix} K^{uu} & 0 \\ 0 & K^{dd} \end{bmatrix}_{t}^{-1} \begin{Bmatrix} R^{u} \\ R^{d} \end{Bmatrix}_{t} \tag{32}$$

in which the tangent stiffness matrices are calculated as follows

$$\begin{cases} K_{ij}^{uu} = \int_{\Omega} [(1-d)^2 + k] (B_i^u)^T \mathbb{C} B_j^u dV \\ K_{ij}^{dd} = \int_{\Omega} [(2\mathscr{H} + \frac{G_c}{l}) N_i N_j + G_c l (B_i^d)^T \omega B_j^d] dV \end{cases} \tag{33}$$

The phase field method mentioned above is implemented using Abaqus UEL subroutine which allows for user-defined computation of the element tangent stiffness matrices and the nodal force vectors. In this

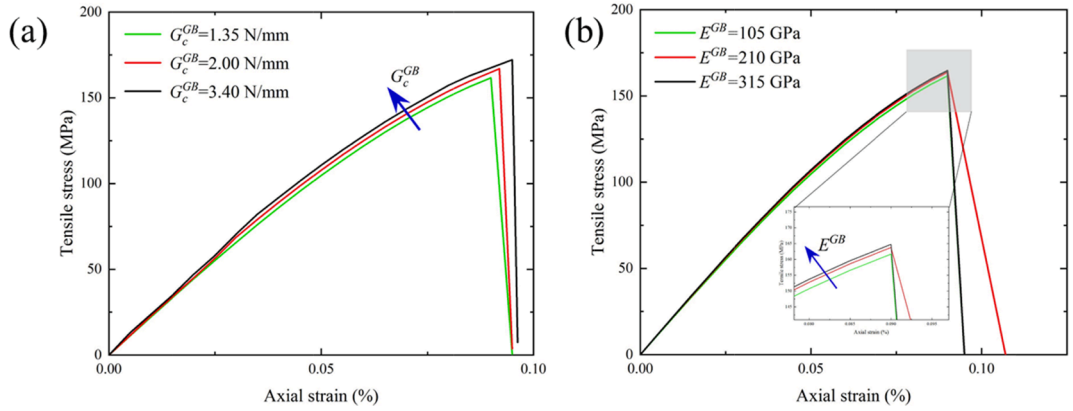


Fig. 8. The stress–strain curves of different models. **a** the effect of grain boundaries critical energy release rate (G_c^{GB}); **b** the effect of grain boundaries Young's modulus (E^{GB}).

paper, 2D simulations with plane strain assumptions were used [30].

3. GB-PFM Modeling procedure

3.1. Model description

The GBM based on the Particle Flow Code is a numerical simulation method for simulating brittle polycrystalline rocks, which microcracks are present as the contact bond between balls break [57]. Inspired by this, in the GB-PFM based on Abaqus, we compose elements into mineral grains and grain boundaries, respectively. The mechanical properties of mineral grains and corresponding grain boundaries are different, such as critical energy release rate, Young's modulus, and Poisson's ratio, *etc*. The grain boundaries in GB-PFM are treated as interconnected networks which represented by elements with quadrilateral meshing. The GB-PFM based on Abaqus characterizes: (a) the grain boundaries is simulated as defects with smaller Young's modulus and critical energy release rate; (b) material heterogeneities of different mineral grains (K-feldspar, plagioclase, quartz and biotite) and (c) intergrain cracking.

For brittle polycrystalline rocks, such as LdB granite, the most important numerical simulation procedure is to qualitatively reconstruct the microstructure. Voronoi tessellation technology has been extensively used in simulating the microstructure of brittle polycrystalline rocks [33,54,92,35,64,60,34]. It is natural to note that Voronoi tessellation is the most common and suitable method to construct the polycrystalline model by means of an assembly of polyhedral [34].

Neper is an open source software for polycrystal generation in 2D or 3D (https://neper.info/), which can generates polycrystals as Voronoi tessellations in a limited domain (like rectangle) [61]. In this work, Neper is used to generate Voronoi tessellations. Considering a set of seed point G_i in 2D domain D ($D \in \mathbb{R}^2$), the Voronoi cell is denoted as

$$C_i = \{P(\boldsymbol{x}) \in D | dist(P, G_i) < d(P, D_j) \forall j \neq i \} \quad (34)$$

where *dist*() is the Euclidean distance and the seeds are randomly located in a specified distribution in domain D.

3.2. Generation of GB-PFM

To elaborate on the modeling procedure of the GB-PFM, a square polycrystalline microstructure with 100 mm × 100 mm domain containing 30 grains is built. The number of grains is the same as that of Nguyen-Thanh et al. [52]. The generation procedure is shown in Fig. 2. The GB-PFM generating procedure based on Abaqus is illustrated as follows: (a) Voronoi tessellations with 30 grains (minerals) generated by Neper (Fig. 2a); (b) the Voronoi tessellations file is saved in a format (*.dxf*) that can be imported into AutoCAD 2007 directly. And then offset unit grain inward by a certain distance to form grain boundaries with a certain thickness (Fig. 2b); as an improvement, in this work, grain boundaries are treated as interconnected networks; (c) importing the microstructure in Fig. 2b into Abaqus as a basis for establishing the GB-PFM (Fig. 2c); (d) finite element meshes by Abaqus (Fig. 2d) and grain boundaries networks composed by elements (Fig. 2e). The grain boundaries are interconnected in polycrystalline materials. Ultimately, GB-PFM model is generated by combining UEL with the GBM based on Abaqus.

In the context of brittle rocks considered at the microscale, a crack can be classified into the following categories [22] (see also Fig. 3): (a) intergrain cracks: Fracture propagating along the grain boundaries; (b) intragrain cracks: Fracture propagation through the grains. The microcracking behavior along the grain boundaries or inside the mineral grains can be well captured by the GB-PFM (Fig. 3).

3.3. Model setup

It was found from previous studies that the mineralogical composition [65] and grain size heterogeneity [55,38] had a great effect on the cracking behavior and mechanical properties of brittle rocks. Therefore, in this work, the mineral composition, mineral content, and mineral grain size distribution will be taken into account in the GB-PFM.

The mineral grain boundaries of the actual granite is rough. Although there are methods to obtain the real microstructure of the rocks through digital image processing (DIP) technique, and to reconstruct the microstructure through the GBM, there are still many limitations [92,24]. At the same time, numerical simulation is also limited by the processing power of the computer. Therefore, when simulating the microstructure of granite at the grain scale, all the current numerical methods have some limitations in accurately simulating the real microstructure of granite, and some necessary simplifications are required [33,54,92,34,64,60]. The grain microstructure simulated in this work is not intended to precisely match that of actual LdB granite but to account for some of its grain boundaries, preferential cleavage planes, mineral composition and content, which are the important characteristics of brittle rocks that cannot be simulated by the traditional phase field methods [58,74]. The exact irregular shapes of grain boundaries and spatial distribution of minerals are not considered in this study.

As an example, Fig. 4 provides a complete generation procedure of a polygon based on the real microstructure of LdB granite. First, we identify the mineral species from the micrograph (Fig. 4a), and then sketch the mineral grain boundaries to obtain the complex irregular polygonal boundaries (Fig. 4b). In simulating the microstructure of polycrystalline rocks, the simulation of grain boundaries between mineral grains in GBM is of vital importance, and some simplifications are

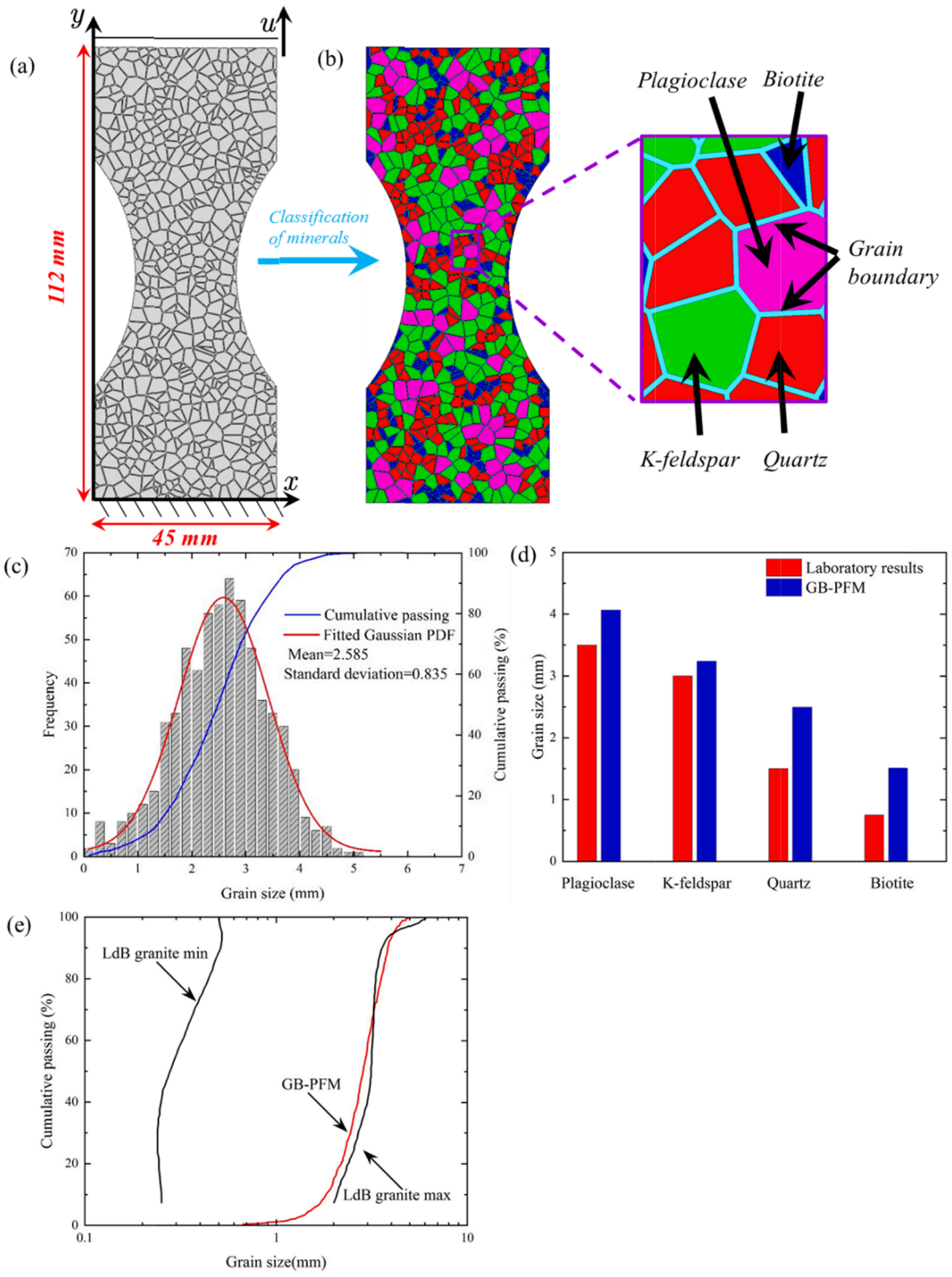


Fig. 9. Numerical GB-PFM model of the LdB granite. **a** Grain structure containing 643 grains, which is generated by Neper; **b** GB-PFM in which polygonal interfaces is replaced by grain boundaries with thickness being equal to 0.2 mm. The colors indicate different minerals. Plagioclase = pink, quartz = red, K-feldspar = green, biotite = blue and Grain boundary = strong cyan. **c** Combined histogram, cumulative passing curve for grains measured in **b** and fitted by the Gaussian probability density function (PDF); **d** Mean grain size of the LdB granite [29] and numerical model in **b**,**e** Grain size distribution of the LdB granite [29] and GB-PFM using Voronoi tessellation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

often required. Second, simplification is performed to obtain a more regular mineral grain boundaries (Fig. 4c). Finally, it was imported into Abaqus to establish finite element model, and the model was meshed to form the GB-PFM (Fig. 4d). It is should be mentioned that the polygon in Fig. 4c is not a Voronoi polygon. The grain shape effect on the mechanical properties and microcracking behavior of brittle rocks, however, is difficult to be studied in laboratory testing because the rock specimen must be destroyed if the internal actual grain shape is to be observed and characterized [55]. In order to simulate the microstructure of polycrystalline rocks, in this work, we will replace the polygon (Fig. 4c) with Voronoi tessellation [34]. It should be noted that this 2D Voronoi tessellation (like Fig. 2a) is only a schematic representation of the grain boundaries of rocks and does not represent the real grain shape characteristics of rocks.

Table 2
Parameters for minerals and grain boundaries in GB-PFM.

	E (GPa)	ν	G_c (N/mm)	β	θ (°)
K-feldspar	69.8	0.28	0.012	10.0	−83.0
Plagioclase	88.1	0.26	0.007	10.0	4.0
Quartz	94.5	0.08	0.068	10.0	−13.0
Biotite	33.8	0.36	0.014	10.0	18.0
Grain boundaries	20.0	0.12	0.005	–	–

4. Comprehensive parametric study

4.1. Selection of calibration parameters

The difficulty of laboratory measurements on the micro- and macro-properties of mineral grains and grain boundaries limits the research of GBM [33,96,34]. In numerical simulations, parameters calibration is an important procedure to match the macroproperties of numerical models to macroproperties of rock specimens measured in the laboratory tests. Most microparameters of rocks are difficult to measure from laboratory tests, especially when the rock heterogeneity increases the grain boundaries and mineral components in a grain-based numerical model, like GBM-UDEC, GB-FDEM, GBM-PFC, and the GB-PFM proposed in this paper. Due to the complexity of the interactions among mineral grain shapes, sizes, microproperties heterogeneity and mechanical properties of different minerals, there is no standard calibration procedure to determine the microparameters.

As mentioned by Li et al. [36], two parts of microparameters are distinguished in GBM: certain microparameters and uncertain micro-parameters. For the GB-PFM proposed in this paper, mean grain size, minerals content, Young's modulus and Poisson's ratio of different minerals have been reported in the previous studies [29,3,33,44] as listed in Table 1. These microparameters are regarded as certain parameters. However, because of the difficulty of distinguishing the grain boundaries and experimentally measuring the mechanical properties of grain boundaries, the microparameters of grain boundary thickness,

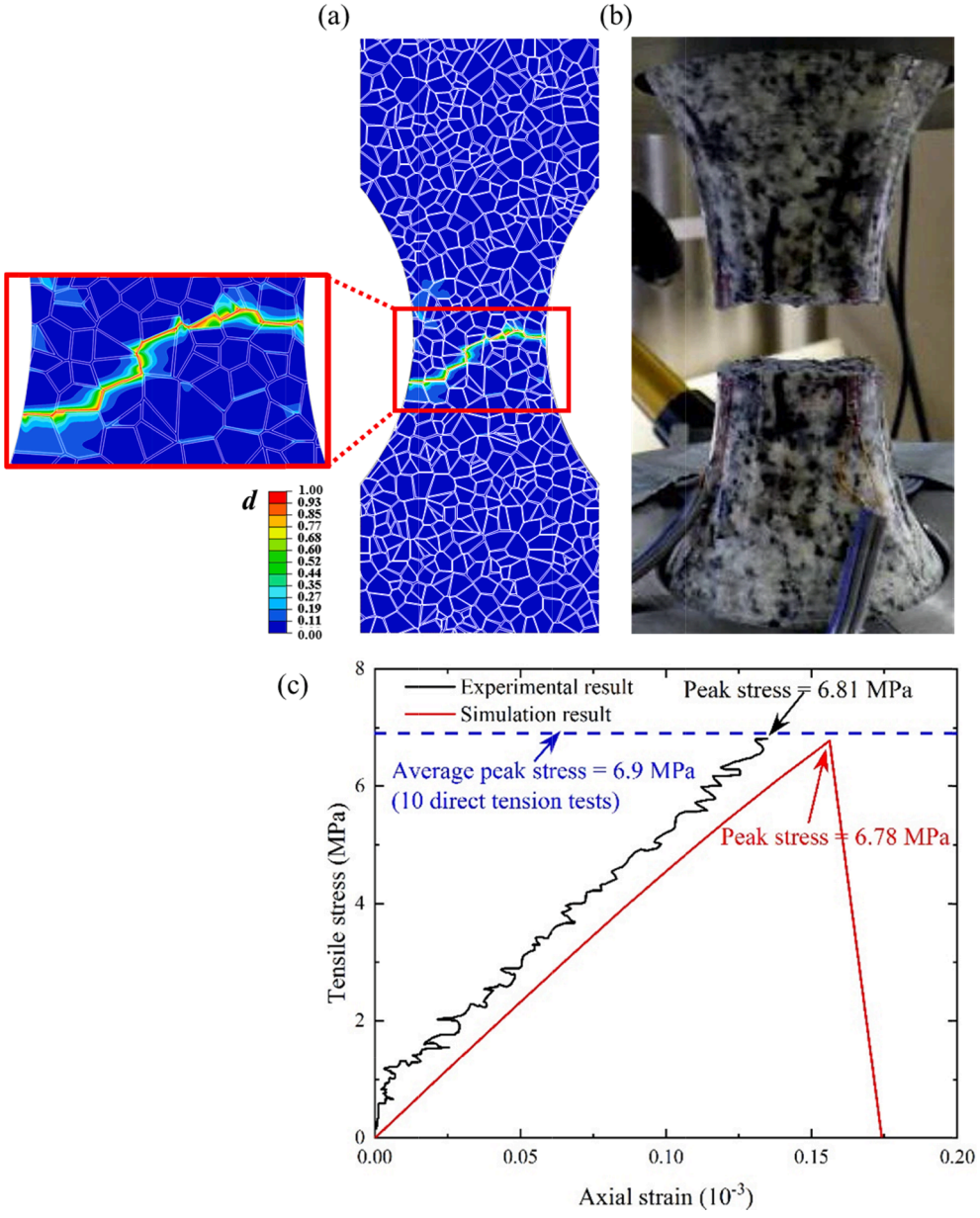


Fig. 10. Numerical and laboratory results of direct tension tests. **a** Numerical macroscopic tensile fracture response; **b** Laboratory result of failure pattern [69] and **c** Stress–strain curves in laboratory tests [46,60] and corresponding numerical result.

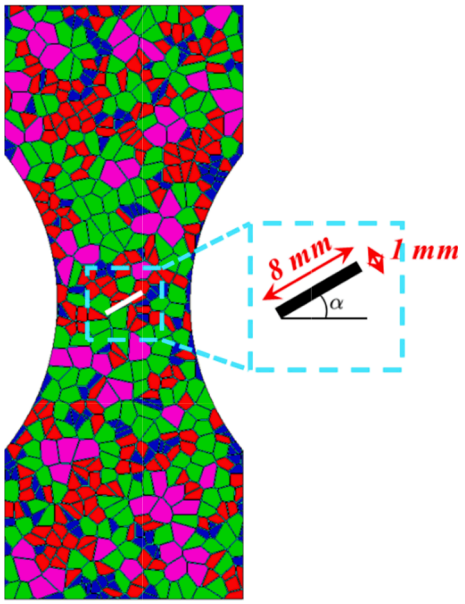


Fig. 11. Geometry of heterogeneous LdB granite with a single pre-existing flaw.

grain boundary critical energy release rate, grain boundary Young's modulus and grain boundary Poisson's ratio are regarded as uncertain parameters. Moreover, the critical energy release rate G_c, which is an important parameter. But its direct measurement by laboratory tests is difficult for rocks, like granites [9]. For microscopic numerical model proposed in this paper, the critical energy release rate of different minerals and grain boundaries are the most important parameters. While the theoretical value of critical energy release rate G^o_c can be inferred from the fracture toughness, such theoretical value is not adopted for this work. However, as mentioned in GBM-PFC by Hofmann et al. [23] and Hu et al. [24], the theoretical value or "exact" value will be used as an important reference value in GB-PFM.

As mentioned above, the microparameters need to be calibrated in GB-PFM are critical energy release rate of four minerals and grain boundaries parameters such as Young's modulus, Poisson's ration, critical energy release rate and thickness. Firstly, the grain boundaries thickness is comprehensively determined based on sensitivity analysis, computer calculation efficiency, and calculation time. It is should be mentioned that in this study, all the grain boundaries parameters are assigned with the same value [55,96,24,34]. It should be noted again that the preferred cleavage orientation θ in different grains are randomly generated.

Therefore, the GB-PFM proposed in this work has a total of seven uncertain parameters including grain boundaries parameters such as Young's modulus, Poisson's ration, critical energy release rate and critical energy release rate of four minerals (Specifically for K-feldspar, quartz, plagioclase and biotite).

4.2. Parametric study

A case of unstable crack propagation in a square with the 100 mm × 100 mm domain containing 30 grains (see also Fig. 2) and an initial crack under direct tension condition (Fig. 5) is proposed. The boundary conditions are shown in Fig. 5a. As illustrate in Fig. 5b, a randomly preferred cleavage plane orientation with each grain.

4.2.1. The effect of grain boundaries thickness (GB_{th})

The effect of grain boundaries thickness on macro mechanical properties and cracking behavior under uniaxial tension is investigated. The effect of grain boundaries thickness are performed for a group of models with 0.1 mm, 0.2 mm, 0.5 mm, 0.8 mm, and 1.0 mm thickness, as illustrated in Fig. 6a. With the characteristic element length being equal to 0.2 mm (see also Fig. 6b), the corresponding quadrilateral elements are 302 891, 299 213, 296 433, 291 121, and 292 485, respectively. We adopt a length scale that is 8 times larger than the characteristic element length to ensure mesh-independent results [30], $l = 1.6$ mm. The input GB-PFM parameters are as follows: (1) mineral grains [30]: $E = 210$ GPa, $\nu = 0.3$, and $G_c = 2.7$ N/mm; (2) grain boundaries: $E = 105$ GPa, $\nu = 0.15$, and $G_c = 1.35$ N/mm. The parameters of the grain boundaries are half of the parameters of the mineral grains.

As illustrate in Fig. 7, the effect of grain boundaries thickness on the cracking behavior and stress–strain curves under uniaxial tension and thin section results observed in laboratory tests. It is seen from Fig. 7a–b that the grain boundaries thickness has a great influence on the macromechanical properties of model, like peak tensile strength (*PTS*) and Young's modulus *E*. It is should be accepted that the *PTS* and *E* is sensitive to grain boundaries thickness. Based on the numerical results, there is a linear correlation between *PTS*, *E* and GB_{th} with the coefficient of determinations $R^2 = 0.981$, 0.988, respectively. An R^2 of 1 indicates that the regression predictions perfectly fit the data. Comparatively speaking, grain boundaries thickness has a more direct impact on macromechanical property *PTS* and *PTS* is increased from 140.4 MPa to 163.2 MPa when the GB_{th} variance decreases from 1.0 mm to 0.1 mm. The influence of grain boundaries thickness on cracking behavior is shown in Fig. 7c–d. A phenomenon for intergrain and intragrain crack patterns is observed in this numerical simulation. The propagation direction of each intragrain crack is different, as it depends on the preferred cleavage plane orientation. Therefore, preferred cleavage plane orientation have a great influence on the propagation path of

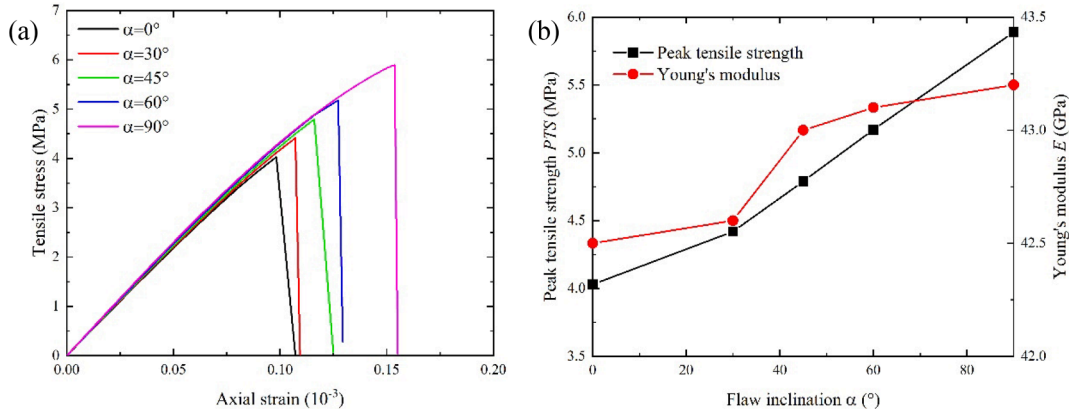


Fig. 12. Mechanical properties of granite with different flaw inclination. **a** Stress–strain curves of five models; **b** Peak tensile strength and Young's modulus with various flaw inclination;

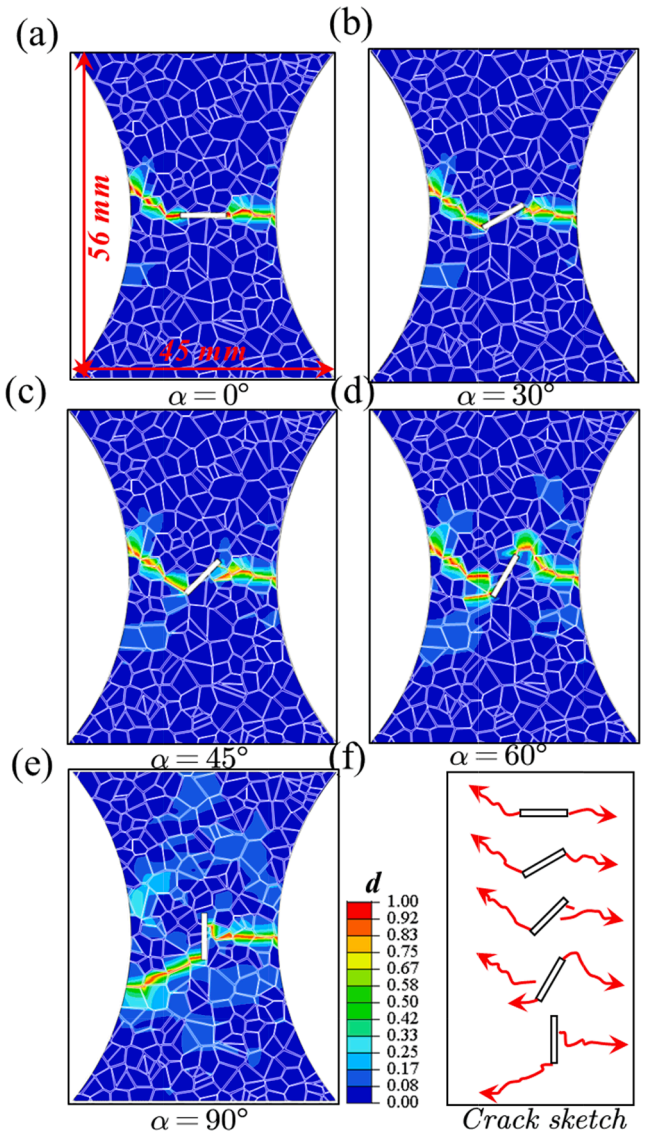


Fig. 13. The development of macroscopic crack responses and crack sketch. **a** $\alpha = 0°$; **b** $\alpha = 30°$; **c** $\alpha = 45°$; **d** $\alpha = 60°$; **e** $\alpha = 90°$; **f** crack sketch of five models.

microcracks. From the failure patters of two specimens (Fig. 7c–d) under uniaxial tension tests, it was observed that with increasing the grain boundaries thickness, the intergrain cracks was increased (Fig. 7d). In model with larger grain boundaries thickness, the path for crack propagation is along the grain boundaries which results in a series of intergrain cracks (Fig. 7d). This route will lose advantages as the model is built with narrower grain boundaries (Fig. 7c). When the microstructure heterogeneities and the material heterogeneities are not considered, that is, when a homogeneous numerical specimen is established, the crack propagation direction will be perpendicular to the direction of tensile stress, and the crack patterns will appear a regular straight line (Fig. 7e) [30]. It is inconsistent with what is observed in the laboratory tests (Fig. 7f) [92]. Conversely, the simulation results based on GB-PFM (Fig. 7c–d) are basically consistent with the crack patterns observed in laboratory tests (Fig. 7f).

In fact, a sufficiently small grain boundaries thickness should be specified to be closer to the real microstructure of polycrystalline rocks. Moreover, with smaller grain boundaries thickness, the characteristic element length is smaller for obtain good quality of elements, and the number of elements will increase exponentially, which results in great reduction in calculation efficiency and a significant increase in calculation time. Note that the number of elements will take more than 2 million with $GB_{th} = 0.05$ mm and the characteristic element length being equal to 0.025 mm. This is very undesirable.

Therefore, the grain boundaries thickness with 0.2 mm is assigned for next simulations to reduce calculation time and improve computational efficiency. The corresponding computation time is approximately 15 h (for specimen in Fig. 7c), which is desirable.

4.2.2. *The effect of grain boundaries critical energy release rate (G_c^{GB}) and grain boundaries Young's modulus (E^{GB})*

Shown in Fig. 8 is the influence of G_c^{GB} and E^{GB} on stress–strain curves. The peak tensile strength is sensitive to G_c^{GB} and E^{GB}. Comparatively speaking, G_c^{GB} has a more impact on peak tensile strength and this value increased from 161.67 MPa to 172.19 MPa when the G_c^{GB} variance increases from 1.35 N/mm to 3.40 N/mm. Moreover, the Young's modulus of numerical specimen is not sensitive to G_c^{GB} and E^{GB}.

4.3. *Calibration of parameters*

Based on the comprehensive discussion and parameter analysis mentioned above, the critical energy release rate of different minerals and grain boundaries are the most important parameter in GB-PFM. But we cannot arbitrarily choose the critical energy release rate of minerals and grain boundaries in GB-PFM approach. While the critical energy release rate can be inferred from the fracture toughness under the assumption of linear elastic fracture mechanics, such assumption is maybe inadequate for this work. Because the LdB granite simulated in this paper is heterogeneous material composed of four minerals with different mechanical properties. And the theoretical value of the critical energy release rate under the plane strain assumption will be used just as an important reference value in GB-PFM. This is an acceptable method to select higher critical energy release rate for quartz and biotite and a lower critical energy release rate for feldspar (K-feldspar or plagioclase). Similar parameter selection rules have been used in other studies, such as Mode I/II fracture energy in GB-FDEM [1,36].

The parameters calibration step for GB-PFM by numerical direct tension tests and matching the macromechanical properties to the laboratory tests. The average values of direct tensile strength (*DTS*) of the LdB granite were 6.9 MPa [46]. Fig. 9a shows the geometry and boundary conditions of 2D dog-bone shaped model under direct tension containing 643 grains, which is generated by Neper [61]. Using the same dimensions (112 mm × 45 mm) as the laboratory test setup [46,66,60]. Refer to Table 1 for the mineral composition and the associated mean grain size of the LdB granite. The plagioclase has the largest grain size, while the grain size of biotite is the smallest. Then, according to the mineral composition and content of the LdB granite, as well as the relationship between the mean grain size of different minerals, specify the type of mineral for each Voronoi grain with the grain boundaries thickness being equal to 0.2 mm (as Fig. 9b shown). The numerical model consists of approximately 20.1% plagioclase, 39.9 % K-feldspar, 29.9% quartz and 10.1% biotite, which is the content obtained after removing the area of the grain boundaries. The mineral composition and content are in accordance with laboratory observations (as Table 1 shown). The histogram, cumulative passing curve for 643 grains are shown in Fig. 9c. The largest and smallest of grain size are 5.07 mm and 0.10 mm, respectively. The mean grain size of the LdB granite [29] is smaller than numerical model generated by GB-PFM (Fig. 9d). For comparison, the grain size distribution of LdB granite [29] and simulated numerical model are shown in Fig. 9e.

The optimal parameters for minerals and grain boundaries as shown in Table 2 after detailed sensitivity analysis for the GB-PFM model calibrated by direct tension tests in laboratory tests. The critical energy release rate are different from the theoretical values for the minerals (see Table 1), because these modeled values in GB-PFM do not represent the whole mineral but one single element within the mineral [23]. However,

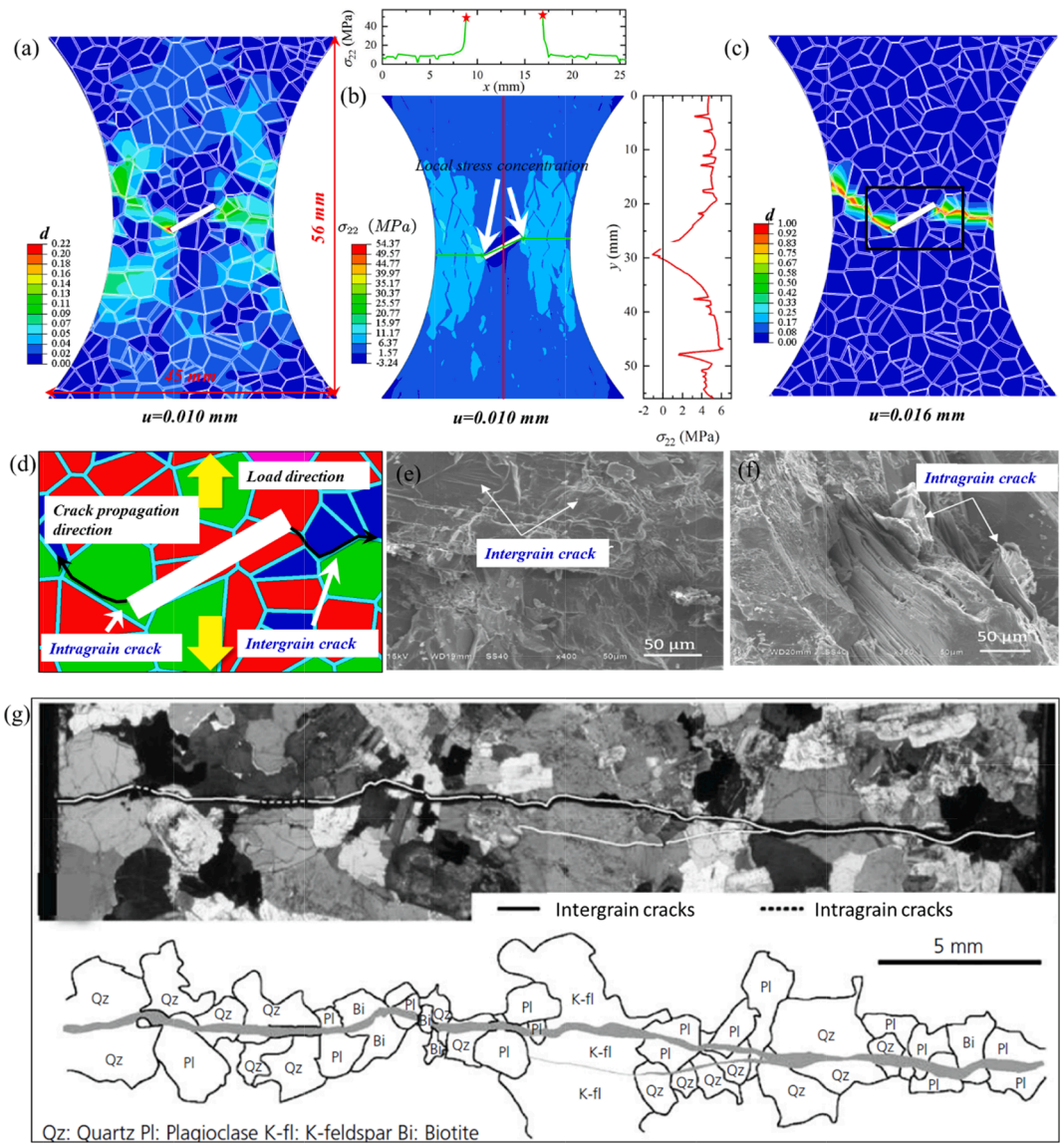


Fig. 14. Typical direct tension test result in numerical simulation ($\alpha = 30°$) and thin section results from laboratory tests. **a** Macroscopic tensile fracture response for displacement of $u = 0.010$ mm; **b** The axial stress distribution in y direction (σ_{22}) and the crossed section of stress at different positions, red line is y direction section, green line is x direction section; **c** Macroscopic tensile fracture response for displacement of $u = 0.016$ mm; **d** Zoomed-in view of the black box in **c**, K-feldspar = green, plagioclase = pink, quartz = red, biotite = blue, Grain boundary = strong cyan, crack propagation direction = black, load direction = yellow; The SEM micrograph shows the microscopic fracture patterns at the crack surface of granite under direct tension tests: **e** Intergrain crack and **f** intragrain crack [90],**g** Photomicrographs of tensile fractures of Inada granite [20] (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the rules of the modeled values and theoretical values are consistent, such as largest of critical energy release rate for quartz. The stress–strain curves and macroscopic tensile fracture response obtained from the experiments and numerical simulations are depicted in Fig. 10. Fig. 10a–b show the macroscopic tensile fracture behavior in laboratory tests and numerical simulations. For direct tension tests, macro tensile failure is observed in laboratory, which is resulted from tensile micro-cracks due to lower tensile strength for brittle materials (Fig. 10b). This tensile fracture is ultimately revealed by numerical simulation. In the GB-PFM model, a tensile diffuse crack bands is observed in the middle of the model (see Fig. 10a). Similar numerical results have also been reported by Peng et al. [54], Saadat and Taheri [64] and Qi et al. [60]. The width of tensile diffuse crack bands is significant influenced by the grain boundaries thickness and the length scale parameter l. The stress–strain behaviors, including *DTS* and Young's modulus, obtained by numerical simulations can follow well with the results obtained by laboratory tests (Fig. 10c). The Young's modulus obtained by laboratory tests is 40.38 GPa [46], the Young's modulus obtain by GB-PFM is 43.30 GPa with an error of 7.2%. The *DTS* obtained by laboratory tests [46] and GB-PFM are 6.9 MPa, 6.78 MPa, respectively, with an error of 1.7%. These results above show that the GB-PFM can reproduce the microcracking behaviors and mechanical properties of the LdB granite under direct tensile condition.

5. Cracking behavior of granite with single pre-existing flaw

Rock mass exhibits obvious discontinuity induced by internal flaws, joints and faults that form during long-term geological history. It is well-known that underground mining, hydropower, and civil engineering are often constructed on or in pre-cracked rock mass. The pre-existing flaws causes stress concentration inside the rock under the action of external load, which reduces the strengths of brittle rocks [13]. In laboratory

tests [86,26,75] and numerical simulations [16,63], studying the strengths and cracking behaviors of granite containing pre-existing flaws has provided a good way to understand the damage evolution of brittle rocks. However, the previous laboratory tests or numerical simulations focused on damage evolution under compressive loading and researchers seldom study the microcracking behavior of pre-cracked granite under tension condition.

In this section, we numerical investigate microcracking behavior and mechanical properties of heterogeneous LdB granite with a single pre-existing unfilled flaw (flaw inclination $\alpha = 0°$, $30°$, $45°$, $60°$, $90°$) under direct tensile loading condition by GB-PFM (Fig. 11). Pre-existing unfilled flaw is formed in Abaqus by removing specific areas of the shell.

The stress–strain curves of heterogeneous LdB granite with a single pre-existing flaw under direct tensile tests is shown in Fig. 12a, and the peak tensile strength and Young's modulus obtained at different flaw inclination illustrated in Fig. 12b.

It was observed that mechanical behavior of all five numerical models with various flaw inclination (flaw inclination $\alpha = 0°$, $30°$, $45°$, $60°$, $90°$) are identical to the intact numerical model (Fig. 10c). However, the *PTS* is greatly reduced when the flaw inclination angle is less than $60°$. The peak tensile strength obtained by specimens with $\alpha = 60°$and intact specimen are 5.17 MPa, 6.78 MPa, respectively, with a reduction of 23.7%. It shows that the existence of pre-existing flaw will greatly reduce the uniaxial tensile strength of brittle rocks.

The macroscopic crack responses and crack sketch are illustrated in Fig. 13. The microcracks initiated from the tip of the flaw and propagated almost perpendicular to the direction of uniaxial tensile loading (Fig. 13). And when $\alpha = 0°$, the macroscopic cracks surface morphology formed almost horizontally (Fig. 13a). In contrast, when $\alpha = 60°$, the macroscopic cracks surface morphology formed an 'S' shape crack (Fig. 13d). When $\alpha = 90°$ the crack tended to propagate from both upper and lower ends of pre-existing flaw (Fig. 13e).

The rock heterogeneity has greatly effect on the cracking behavior of brittle rocks at the grain scale [55,35]. However, in the previous PFMs for simulating the cracking behavior of rocks or rock-like materials, the grain-scale heterogeneity is rarely considered [91,58,87]. Typical direct tension test result in numerical simulation ($\alpha = 30°$) and thin section results from laboratory tests are plotted in Fig. 14 and grain boundaries networks are displayed with white lines. Macroscopic tensile fracture response for displacement of $u = 0.010$ mm is shown in Fig. 14a. From a microview of heterogeneous stress field distribution, the microstructure heterogeneities and material heterogeneities causes local stress concentration at grain boundaries and tips of pre-existing flaw (see Fig. 14b). The local stress concentration is the major reason that local failure and leads to microcrack initiation [13,36]. The maximum axial stress distribution in y direction (σ_{22}) is 54.37 MPa, which appears at both upper and lower ends of pre-existing flaw (Fig. 14b). The failure pattern of the model ($\alpha = 30°$) in direct tension test ($u = 0.016$ mm) is presented in Fig. 14c. A tensile macrocrack characterized by horizontal tensile microcracks with a vertical angle relative to the tensile axis cutting through the numerical model (Fig. 14c). The numerical simulation results can better reveal the failure patterns obtained in the experiments [90]. The numerical simulation results are in good agreement with others [64]. Fig. 14d shows the crack propagation path from the black box in Fig. 14c. The result indicate that the macroscopic tensile cracks are composed of intragrain crack and intergrain crack. Because the grain boundaries critical energy release rate (G_c^{GB}) is lower than that of other mineral grains, microcracks tend to grow along the grain boundary during the propagation processes, resulting in smoother intergrain cracks. At the same time, even when there is a preferred cleavage plane inside the minerals, there are still more choices of microcrack propagation paths than the grain boundary, which makes the intragrain cracks rougher and not smooth. Fig. 14e–f show the SEM micrograph of Mengdigou granite under direct tension test of intergrain crack, intragrain crack, respectively [90]. The SEM results (Fig. 14e–f) indicated that the macroscopic tensile failure of granite can be considered an integration of microcracks. In addition, the laboratory tests results of Fujii et al. [20] supported the numerical simulation results and proved that the macroscopic tensile cracks are composed of intergrain and intragrain cracks at the grain scale (Fig. 14g). In summary, the failure patterns of pre-cracked granite observed from laboratory tests can be revealed and predicted by the GB-PFM.

Compared to the others PFMs [91,87,74], which is not take into account microstructure of brittle rocks, where microcracks are smooth and straight. If no improvement is made, the PFMs cannot effectively simulate the microstructure of the brittle rocks. This is the most important motivation of improving PFM by coupling GBM to taken into account the grain structure of brittle rocks. Therefore, microcracks in GB-PFM have a variety of relatively tortuous propagation paths. Due to the existence of material heterogeneities of different mineral components and microstructure heterogeneities, the path of the macrocracks obtained from GB-PFM shows more noticeable tortuous nature as compared with that results by others PFMs [91,87,74]. In addition, the stress field distribution obtained by GB-PFM is agree well with the theoretical results [80,47,13] and numerical simulations [79,40,41]. Therefore, the GB-PFM proposed in this paper is very suitable to simulate the complex crack propagation behavior of brittle rocks, which not only greatly expands the application of phase field method, but also provides a new perspective for deeper understanding of the damage evolution and crack propagation of brittle rocks.

6. Discussion and limitations

Although the proposed GB-PFM is an effective approach that considers the realistic microstructure of brittle rocks in an explicit manner and goes beyond the traditional PFMs in capturing the material heterogeneities of different mineral components, grain boundaries networks and intragrain fracturing behavior, it still has some limitations, which should be overcome in future work and are discussed in the following.

GB-PFM is more suitable to reproduce the material and microstructure heterogeneities of rocks as compared with the traditional PFMs, one limitation in the modeling is that each mineral grain is homogeneous. Although we considered the preferred cleavage plane inside the mineral grain, it is far from enough. In natural rocks, there is also heterogeneous within a single crystal mineral [15]. Such as strength anisotropy in single quartz crystal mineral [19,78]. But it remains difficult to take the microstructure and strength heterogeneity of single crystal minerals into account by current computer technology and numerical techniques, such as PFMs.

Despite the work in calibration of GB-PFM, there is no standard method to determine microparameters in grain based model. In most cases, the microparameters for grains and grain boundaries are calibrated by trial and error. Even if the GB-PFM model shows the expected mechanical properties and failure patterns under direct tension tests, but how each microparameter is truly assigned has not been solved. So, a series of case studies should be performed to establish empirical laws. At the same time, we can establish a microparameter prediction model by using machine learning algorithms [82,17] in the future work.

For laboratory scale, the scale level is millimeter (mm). However, for engineering scale, like tunnels or other huge models, the scale level is meter (m). The research results show that the mechanical properties of brittle rocks have scale effect [68,39,56]. It should be noted that the proposed GB-PFM in this work maybe only suitable for laboratory scale. For larger size models, like tunnels, the exploration of a more detailed study of GB-PFM is still warranted in the future work.

In addition, we only study the mechanical properties and microcracking behavior under direct tension in this work. However, rocks may also be loaded by compression, shear, or mixed mode. At present, some new methods have been reported to simulate the compressive, shear fracture or mixed mode fracture based on phase field method, such as

presenting a new driving force [94] or developing a double-phase-field formulation [87,18]. To combine the GB-PFM and the method of simulating compressive, shear fracture or mixed mode fracture will be an interesting research topic.

7. Conclusions

To characterize behavior of intragrain fracturing within minerals and provide a more realistic representation of brittle rocks microstructures, grain-based model is developed in PFM, an extended GB-PFM, which models crystalline rocks as an assembly of Voronoi tessellations. The grain scale structure such as grain size distribution, mineral composition and mean grain size is realistically reproduced based on laboratory tests. The grain boundaries between minerals are assumed as elements with smaller Young's modulus and critical energy release rate allowing breakable and deformable. Specifically, grain boundaries properties, such as grain boundaries thickness and critical energy release rate, have greatly influences on the microcracking behavior and mechanical properties of brittle rocks. Greater grain boundaries thickness results in lower *DTS* and lower Young's modulus, and a more number of the intergrain cracks.

Five models with single pre-existing flaw (flaw inclination $\alpha = 0°$, $30°$, $45°$, $60°$, $90°$) are examined based on GB-PFM under direct tension. Microparameters calibration is performed through matching the mechanical properties and failure patterns with the laboratory tests. Using the GB-PFM, a considerable agreement of LdB granite both in terms of stress–strain curves and microcracking behavior under direct tension are achieved. Compared with the intact model, the existence of pre-existing flaw greatly reduces the *DTS* and Young's modulus of the pre-cracked model. A tensile macrocrack characterized by horizontal tensile microcracks with a vertical angle relative to the tensile axis cutting through the numerical model ($\alpha = 30°$). Moreover, the path of the macrocracks obtained from GB-PFM shows obvious tortuous nature as compared with the results of traditional PFMs. Additionally, the variation of flaw inclination angles has a great influence on microcrack initiation and propagation at the grain scale. From a microscopic point of view, the failure mode obtained by GB-PFM reveals the macroscopic cracks produced by pre-existing flaw, which are composed of intragrain cracks and intergrain cracks. Moreover, the stress field distribution obtained by GB-PFM is agree well with the theoretical results and other numerical simulations.

Our work proves that GB-PFM can well simulate the mechanical properties and microcrack propagation behavior of brittle rocks. The method proposed in this paper can also be extended to the study of the progressive fracture processes of other brittle rocks such as sandstone and marble, which greatly expands the application of the PFM in brittle rocks. Therefore, GB-PFM is a numerical simulation method with great potential in rock mechanics and rock engineering.

CRediT authorship contribution statement

Xunjian Hu: Conceptualization, Methodology, Software, Resources, Investigation, Formal analysis, Visualization, Writing – original draft. **Xiaonan Gong:** Funding acquisition, Supervision. **Ni Xie:** Writing – review & editing. **Qizhi Zhu:** Writing – review & editing, Software. **Panpan Guo:** Writing – review & editing, Software. **Haibo Hu:** Writing – review & editing. **Junjie Ma:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to the National Natural Science Foundation of China (51778575) for the financial support. The first author acknowledges Emilio Martínez Pañeda, Imperial College London, for his help in phase field method numerical simulation. The first author acknowledges Yujie Huang, North University of China, for his help in formal analysis.

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